4.1 Topological Spaces

References:

- General Topology [Kel71] (Chap. 1)
- General Topology [Wil70]

Topology is useful in algebraic geometry, but often the natural topologies are usually much coarser so the theory looks rather different.

Definition 4.1.1 (Topological Space)
A topological space $(X, \mathcal{T}_X)$ consists of a set X and family of open sets $\mathcal{T}_X \subseteq \mathcal{P}(X)$ satisfying the following properties

- $X, \varnothing \in \mathcal{T}_X$
- $U_i \in \mathcal{T}_X \implies \bigcup_{i \in I} U_i \in \mathcal{T}_X$
- $U, V \in \mathcal{T}_X \implies U \cap V \in \mathcal{T}_X$

A subset $Z \subset X$ is said to be closed iff $X \setminus Z$ is open. We may equivalently define the topology in terms of closed sets.

Proposition 4.1.2
Let X be a topological space. Both the open sets and closed sets form a distributive lattice under inclusion.

Definition 4.1.3 (Subspace topology)
*Let $Y \subset X$, then we may define the **subspace topology** on Y by*

$$\mathcal{T}_Y := \{U \cap Y \mid U \in \mathcal{T}_X\}$$

when Y is open then this is given by

$$\mathcal{T}_Y = \{U \subseteq Y \mid U \in \mathcal{T}_X\}$$

Definition 4.1.4 (Base)
We say $\mathcal{B} \subseteq \mathcal{P}(X)$ is a base (of open sets) on X if

- *For every $x \in X$ there is a $U \in \mathcal{B}$ such that $x \in U$*
- *Suppose $U, V \in \mathcal{B}$ and $x \in U \cap V$ then there exists $W \in \mathcal{B}$ such that $x \in W \subseteq U \cap V$*

Proposition 4.1.5 (Topology generated by a base)
Let $\mathcal{B}$ be a base, then the following is a topology on X

$$\mathcal{T}_{\mathcal{B}} := \left\{ \bigcup_{U_i \in I} U_i \;\middle|\; I \subseteq \mathcal{B} \right\}$$

i.e. the set of arbitrary unions of sets in $\mathcal{B}$.

Proposition 4.1.6 (Base generating topology)
A base $\mathcal{B}$ satisfies $\mathcal{T}_{\mathcal{B}} = \mathcal{T}_X$ if and only if

a) $\mathcal{B} \subset \mathcal{T}_X$
b) For every $x \in U$ and $U \in \mathcal{T}_X$ there exists $V \in \mathcal{B}$ such that $x \in V \subseteq U$.

In this case we say $\mathcal{B}$ is a base for (the topology on) X.

Proposition 4.1.7 (Subbase for a topology)
Let X be a set and $\mathcal{B}$ be an arbitrary family of subsets. Then there exists a topology $\mathcal{T}_{\mathcal{B}}$ such that

a) $\mathcal{B} \subset \mathcal{T}_{\mathcal{B}}$
b) For every topology $\mathcal{T}'$ containing $\mathcal{B}$ we have $\mathcal{T}_{\mathcal{B}} \subset \mathcal{T}'$.

*We say that $\mathcal{B}$ is a **subbase** for the topology $\mathcal{T}_{\mathcal{B}}$. Explicitly $\mathcal{T}_{\mathcal{B}}$ consists of arbitrary unions of finite intersections of elements of $\mathcal{B}$.*

Proposition 4.1.8 (Base for subspace topology)
Let $(X, \mathcal{T})$ be a topological space, and $Y \subset X$ has the subspace topology $\mathcal{T}_Y$ (4.1.3). If $\mathcal{B}$ is a base for $(X, \mathcal{T})$ then

$$\mathcal{B}_Y := \{U \cap Y \mid U \in \mathcal{B}\}$$

is a base for $(Y, \mathcal{T}_Y)$.

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Definition 4.1.9 (Adherent point)
*For $Y \subset X$ we say x is an **adherent point** of Y if every open neighbourhood of x intersects Y.*

*Similarly we say x is a **limit point** (or **accumulation point**) for Y if every open neighbourhood of x intersects Y at a point other than x.*

Proposition 4.1.10 (Closed point)
Let $x \in X$ then TFAE

- *$\{x\}$ is closed*
- *For every $y \neq x$ there is $U \ni y$ such that $x \notin U$.*

Definition 4.1.11
For a topological space X let X_0 denote the subset of closed points.

Definition 4.1.12 (Open Embedding)
*Let $f : X \to Y$ be a map of topological spaces. We say it is an **open embedding** if one of the following equivalent conditions holds*

a) $f(X)$ is open and f is a homeomorphism onto its image
b) f is continuous, open and injective.

4.1.1 Axioms of Countability

Definition 4.1.13 (Neighbourhood)
*Let X be a topological space and $x \in X$. We say a set $V \subset X$ is a **neighbourhood** of x if $x \in U \subset V$ for some open set U. We write $\mathcal{U}_x$ for the neighbourhoods of x, and $\mathcal{O}_x$ for the open neighbourhoods of x.*

*We say that a collection of neighbourhoods $\mathcal{B}_x$ is a **local base** for x if every neighbourhood $U \in \mathcal{U}_x$ contains some neighbourhood $V \in \mathcal{B}_x$. For example $\mathcal{O}_x$ is a local base.*

Definition 4.1.14 (First-Countable)
*Let X be a topological space. X is **first countable** if every point $x \in X$ has a countable local base.*

Definition 4.1.15 (Second-Countable)
*Let X be a topological space. X is **second countable** if there exists a countable basis. Evidently this is stronger than first countability.*

Example 4.1.16
For example a metric space is always first countable as we may consider balls of radius $\frac{1}{n}$.

Further $\mathbb{R}$ is second countable as we may consider open intervals with rational endpoints.

4.1.2 Closure

Definition 4.1.17 (Adherent Point)
*Let X be a topological space and $Y \subset X$. We say that $x \in X$ is an **adherent point** of Y, if $U \in \mathcal{U}_x \implies U \cap Y \neq \varnothing$.*

*We say that x is a **sequential limit point** of Y if there exists a sequence $x_n \in Y$ such that $x_n \to x$.*

Proposition 4.1.18 (Adherent Point)
Let X be a topological space. Then every sequential limit point is an adherent point. If X is first countable the converse holds.

Proof. The first implication is straightforward. For the latter, consider a countable local base at x given by U_n. We may assume wlog that it is decreasing. Then by assumption we may choose $x_n \in U_n \cap Y$. For any $V \in \mathcal{B}_x$ we have $U_N \subset V$ for some N, and therefore $n \geq N \implies x_n \in U_n \implies x_n \in V$. Therefore $x_n \to x$. ∎

Proposition 4.1.19 (Topological Closure)
Let $Y \subset X$ then the following equality holds

$$\bigcap_{\substack{Z \supseteq Y \\ Z \text{ closed}}} Z = \{x \in X \mid x \text{ adherent point of } Y\}$$

*We denote this by $\overline{Y}$ (or $\mathrm{cl}_X(Y)$ to emphasise the ambient space X) and refer to it as the **closure** of Y in X. Furthermore the following properties hold*

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a) $Y \subseteq \overline{Y}$ *and* $\overline{Y}$ *is closed*
b) $Y = \overline{Y}$ *if and only if* Y *is closed*
c) $\left(Y \cap U \neq \varnothing \iff \overline{Y} \cap U \neq \varnothing\right)$ *for any* U *open*

When X *is first countable then* $\overline{Y}$ *is the set of sequential limit points of* Y*.*

Proof. Suppose Z is a closed set containing Y and x is an adherent point of Y. Then $x \notin Z \implies x \in X \setminus Z \implies (X \setminus Z) \cap Y \neq \varnothing$ a contradiction. Conversely assume $x \notin \overline{Y}$ then there exists $Z \supseteq Y$ closed such that $x \notin Z \implies x \in X \setminus Z$. This means x is not an adherent point.

a) An arbitrary intersection of closed sets is closed
b) This follows because $\overline{Y}$ is the smallest closed superset.
c) One implication is clear because $Y \subseteq \overline{Y}$. Conversely if $x \in \overline{Y} \cap U$ then x must be a limit point of Y hence $U \cap Y \neq \varnothing$ as required.

■

Corollary 4.1.20
Let X *be a first countable topological space. Then* $\mathrm{cl}_X(Y)$ *is the set of sequential limit points of* Y*.*

Proposition 4.1.21 (Closure in Subspace Topology)
Let X *be a topological space and* Y *a subset. For* $Z \subset Y$ *we have*

$$\mathrm{cl}_Y(Z) = \mathrm{cl}_X(Z) \cap Y$$

Proof. By (4.1.19) $\mathrm{cl}_X(Z)$ is a closed subset of X and therefore $\mathrm{cl}_X(Z) \cap Y$ is a closed subset of Y in the subspace topology. Therefore $\mathrm{cl}_Y(Z) \subset \mathrm{cl}_X(Z) \cap Y$ by minimality. Similarly $\mathrm{cl}_Y(Z) = W \cap Y$ for W closed in X. By minimality $\mathrm{cl}_X(Z) \subset W \implies \mathrm{cl}_X(Z) \cap Y \subset W \cap Y = \mathrm{cl}_Y(Z)$. ■

Proposition 4.1.22
Let X *be a topological space with* $X = \bigcup_{i \in I} U_i$ *an open cover. Let* $Y \subset X$ *be a subset then* Y *is closed in* X *if and only if* $Y \cap U_i$ *is closed in* U_i *for all* $i \in I$*.*

Proof. One direction is obvious by definition of the subspace topology. Conversely $U_i \setminus (Y \cap U_i)$ is open in U_i and therefore in X. We see that

$$X \setminus Y = \bigcup_{i \in I} U_i \setminus (Y \cap U_i)$$

is open and therefore Y is closed. ■

Proposition 4.1.23
Let $Y_1, ..., Y_n$ *be subsets of a topological space* X*. Then*

$$\overline{Y_1 \cup ... \cup Y_n} = \overline{Y_1} \cup ... \cup \overline{Y_n}$$

Proof. By induction it is sufficient to demonstrate the case $n = 2$. $\overline{Y_1 \cup Y_2}$ is a closed set containing both Y_1 and Y_2. Therefore by definition

$$\overline{Y_1} \cup \overline{Y_2} \subset \overline{Y_1 \cup Y_2}$$

On the other hand $\overline{Y_1} \cup \overline{Y_2}$ is closed and contains $Y_1 \cup Y_2$. Therefore the reverse inclusion follows immediately. ■

Proposition 4.1.24 (Dense subset)
Let $Y \subset X$ *then the following are equivalent*

a) $\overline{Y} = X$
b) $Y \cap U \neq \varnothing$ *for any* $U \subset X$ *open*

In this case we say Y *is* ***dense****.*

Proof. *a)* $\implies$ *b)* Follows from (4.1.19).c)

b) $\implies$ *a)*. Suppose $Y \subseteq \overline{Y} \subsetneq X$ then $X \setminus \overline{Y}$ is an open set not intersecting Y a contradiction. ■

4.1.3 Continuous Maps

Proposition 4.1.25 (Continuous at a point)
Let $f : X \to Y$ *a map of topological spaces and* $x \in X$*. Then the following are equivalent*

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a) $V \in \mathcal{U}_{f(x)} \implies f^{-1}(V) \in \mathcal{U}_x$
b) $V \in \mathcal{B}_{f(x)} \implies f^{-1}(V) \in \mathcal{U}_x$
c) $V \in \mathcal{U}_{f(x)} \implies \exists U \in \mathcal{U}_x$ s.t. $f(U) \subseteq V$
d) $V \in \mathcal{B}_{f(x)} \implies \exists U \in \mathcal{B}_x$ s.t. $f(U) \subseteq V$

where $\mathcal{B}_{f(x)}$ is any local base at $f(x)$ and $\mathcal{B}_x$ is any local base at x. We say that f is ***continuous*** *at x.*

Proof. *a)* $\implies$ *b)* is obvious. Suppose *b)* holds and $V \in \mathcal{U}_{f(x)}$ then by definition there exists $V' \in \mathcal{B}_{f(x)}$ such that $V' \subset V$. By hypothesis $f^{-1}(V') \in \mathcal{U}_x$ from which it follows that $f^{-1}(V) \in \mathcal{U}_x$.

a) $\implies$ *c)* $f(f^{-1}(V)) \subseteq V$

c) $\implies$ *d)* By definition there exists $U' \in \mathcal{B}_x$ such that $U' \subseteq U \implies f(U') \subseteq f(U) \subseteq V$

d) $\implies$ *b)* Observe that $f(U) \subseteq V \implies U \subseteq f^{-1}(V)$ whence $f^{-1}(V) \in \mathcal{U}_x$ as required. ∎

Proposition 4.1.26
Let $f : X \to Y$ and $g : Y \to Z$ be maps of topological spaces such that f is continuous at x and g is continuous at $f(x)$. Then $g \circ f$ is continuous at x.

Proposition 4.1.27 (Characterisation of Continuous Maps)
Let $f : X \to Y$ be a map of topological spaces. Then the following are equivalent

a) *f is continuous at all points $x \in X$*
b) *$V \subset Y$ open $\implies f^{-1}(V) \subset X$ is open*
c) *$Z \subset Y$ closed $\implies f^{-1}(Z) \subset X$ is closed*

Let $\mathcal{B}_1, \mathcal{B}_2$ be bases of open sets for X and Y respectively. Then this is equivalent to the following condition

$$\forall V \in \mathcal{B}_2, \quad \text{and } x \in f^{-1}(V), \quad \exists U \in \mathcal{B}_1 \text{ s.t. } x \in U \subset f^{-1}(V)$$

Proposition 4.1.28
Let $f : X \to Y$ and $g : Y \to Z$ be maps of topological spaces such that f is continuous and g is continuous. Then $g \circ f$ is continuous.

Proposition 4.1.29
Let $f : X \to Y$ be a map of topological spaces and $X = \bigcup_{i \in I} U_i$ an open cover. Then f is continuous iff $f|_{U_i}$ is continuous for all $i \in I$.

Proof. Observe that $f^{-1}(V) = \bigcup_{i \in I} f^{-1}(V) \cap U_i = \bigcup_{i \in I} \left(f|_{U_i}\right)^{-1}(V)$. ∎

Proposition 4.1.30
Let $f : X \to Y$ be a map of topological spaces. Then f is continuous iff

$$f\left(\overline{Z}\right) \subset \overline{f(Z)}$$

for all subsets $Z \subset X$.

Proof. Suppose f is continuous and $f(Z) \subset W$ is closed. Then by (4.1.27) $Z \subset f^{-1}(W)$ is closed. By definition $\overline{Z} \subset f^{-1}(W) \implies f\left(\overline{Z}\right) \subset f(f^{-1}(W)) \subset W$. As W was an arbitrary closed set containing $f(Z)$ we find by definition $f\left(\overline{Z}\right) \subset \overline{f(Z)}$.

Conversely suppose $W \subset Y$ is closed and $Z \coloneqq f^{-1}(W)$. Then $f\left(\overline{Z}\right) \subset \overline{f(Z)} = \overline{W \cap \mathrm{Im}(f)} \subset W \implies f^{-1}\left(f\left(\overline{Z}\right)\right) \subset Z \implies \overline{Z} \subset Z \implies \overline{Z} = Z$ and therefore Z is closed. Therefore f is continuous by (4.1.27). ∎

4.1.4 Quasi-Homeomorphism

Proposition 4.1.31 (Quasi-Homeomorphism)
Let $f : X \to Y$ be a map. The following are equivalent

a) *The map $W \to f^{-1}(W)$ is a bijection between open sets*
b) *The map $Z \to f^{-1}(Z)$ is a bijection between closed sets*

In this case we say f is a ***quasi-homeomorphism****. Such a map induces a bijection between irreducible sets, respectively irreducible components.*

Proof. This follows by taking complements and observing $X \setminus f^{-1}(A) = f^{-1}(Y \setminus A)$. ∎

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Proposition 4.1.32 (Induced Quotient)
Let $\pi : X \to Y$ be a surjective map where X is a topological space. The family

$$\mathcal{T}_Y := \{U \subset Y \mid \pi^{-1}(U) \in \mathcal{T}_X\}$$

is a well-defined topology on Y. Suppose that $\pi^{-1}(\pi(V)) \subset V$ for all V open (resp. closed) subsets of X. Then π is both open, closed and a quasi-homeomorphism.

Proof. As arbitrary intersections and unions commute with inverse image we see that $\mathcal{T}_Y$ is a well-defined topology.

As π is surjective we have $\pi(\pi^{-1}(U)) = U$ for all open subsets $U \subset Y$. Generically $V \subset \pi^{-1}(\pi(V))$ and by assumption the reverse inclusion holds. Therefore we see π induces a quasi-homeomorphism. This also shows π is open and therefore closed. ∎

4.1.5 Kolmogorov Spaces

Definition 4.1.33
*Let X be a topological space. We say that X is **Kolmogorov** or T_0 if for all distinct pairs $x, y \in X$ there exists an open set U such that either $x \in U$ and $y \notin U$, or $x \notin U$ and $y \in U$.*

Proposition 4.1.34 (Indistinguishable points)
Let X be a topological space and $x, y \in X$. The following are equivalent

a) $\overline{\{x\}} = \overline{\{y\}}$
b) $x \in \overline{\{y\}}$ and $y \in \overline{\{x\}}$
c) Every closed $Z \subset X$ contains both or neither of x, y
d) Every open $U \subset X$ contains both or neither of x, y

*We say x, y are **topologically indistinguishable**, otherwise we say x and y are **distinguishable**.*

X is Kolmogorov iff every pair of points are distinguishable.

*Being indistinguishable determines an equivalence relation on X, and we denote by $X_0 := X/\sim$ the **Kolmogorov Quotient**.*

Proof. $a) \implies b)$ Clear

$b) \implies a)$ By definition of closure $x \in \overline{\{y\}} \iff \overline{\{x\}} \subset \overline{\{y\}}$.

$c) \iff d)$ Follows by taking complements

$c) \implies b)$ Suppose $x \in Z$ for Z closed then by hypothesis $y \in Z$. As Z was arbitrary we see that $y \in \overline{\{x\}}$, and symmetrically $x \in \overline{\{y\}}$.

$b) \implies c)$ By assumption $x \in Z \implies y \in Z$ for Z closed and symmetrically $y \in Z \implies x \in Z$. ∎

Definition 4.1.35 (Kolmogorov Quotient)
*Let X be a topological space. The set X_0 of equivalence classes of topologically indistinguishable points is the **Kolmogorov Quotient***

Proposition 4.1.36 (Kolmogorov Quotient Properties)
Let X be a topological space and X_0 be the Kolmogorov Quotient with quotient map $\pi : X \to X_0$. Relative to the induced topology

$$\mathcal{T}_{X_0} := \{W \subset X_0 \mid \pi^{-1}(W) \in \mathcal{T}_X\}$$

π is a quasi-homeomorphism and an open and closed map. Further X_0 is Kolmogorov.

Proof. By (4.1.32) it is sufficient to show that $\pi^{-1}(\pi(U)) \subset U$ for all open subsets $U \subset X$. Suppose $x \in \pi^{-1}(\pi(U))$ then $\pi(x) = \pi(y)$ for some $y \in U$. Then by definition of π, x and y are indistinguishable and so $x \in U$.

Suppose $\pi(x) \neq \pi(y)$ then by definition there exists an open set $U \subset X$ such that $x \in U$ and $y \notin U$. As π is open then $\pi(U)$ is open. By construction $\pi(x) \in \pi(U)$ and $\pi(y) \notin \pi(U)$. This shows that X_0 is Kolmogorov. ∎

Proposition 4.1.37 (Kolmogorov Quotient Functoriality)
Let $f : X \to Y$ be a continuous map, then there is a unique (continuous) map $f_0 : X_0 \to Y_0$ such that the following diagram commutes

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$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X_0 & \xrightarrow{f_0} & Y_0 \end{array}$$

Proof. Given $x, x' \in X$ and suppose $f(x) \nsim f(x')$. Then there exists an open set U such that $f(x') \in U$ and $f(x) \notin U$. Therefore $x' \in f^{-1}(U)$ and $x \notin f^{-1}(U)$ and $x \nsim x'$. Equivalently $x \sim x' \implies f(x) \sim f(x')$. Therefore the map

$$f_0([x]) = [f(x)]$$

is well-defined.

For $W_0 \subset Y_0$ we have $\pi^{-1}(f_0^{-1}(W_0)) = f^{-1}(\pi^{-1}(W_0))$ is open. As π is surjective and open $f_0^{-1}(W) = \pi(\pi^{-1}(f_0^{-1}(W_0)))$ is open, and therefore f_0 is continuous.

Any such f_0 satisfies $\{f_0(x_0)\} = \pi(f(\pi^{-1}(\{x_0\})))$ and so is unique. ∎

Proposition 4.1.38 (Bijections with Homeomorphic Kolmogorov Quotients are Homeomorphisms)
Let $f : X \to Y$ be a continuous bijection of topological spaces such that the induced map $f_0 : X_0 \to Y_0$ (4.1.37) is a homeomorphism. Then f is a homeomorphism.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $\pi_X : X \to X_0$, $\pi_Y : Y \to Y_0$ for the quotient maps, so $\pi_Y \circ f = f_0 \circ \pi_X$. For $x, x' \in X$ we may show

$$x \sim x' \iff f(x) \sim f(x')$$

For $x_0 \in X_0$ with $x_0 = \pi_X(x)$ we have

$$f(\pi_X^{-1}(x_0)) = \{f(x') \mid x' \sim x\} = \{f(x') \mid f(x') \sim f(x)\} = \pi_Y^{-1}(f_0(x_0))$$

where the last equality requires surjectivity of f.

It follows that f is open: for $W \subset X$ open, indistinguishable points lie in exactly the same open sets, so $x \in W$ and $x \sim x'$ forces $x' \in W$; that is, W is a disjoint union of the equivalence classes it meets, which are precisely the fibres of π_X (4.1.35), $W = \bigsqcup_{x_0 \in \pi_X(W)} \pi_X^{-1}(x_0)$, so

$$f(W) = \bigcup_{x_0 \in \pi_X(W)} f(\pi_X^{-1}(x_0)) = \bigcup_{x_0 \in \pi_X(W)} \pi_Y^{-1}(f_0(x_0)) = \pi_Y^{-1}\left(\bigcup_{x_0 \in \pi_X(W)} \{f_0(x_0)\} \right) = \pi_Y^{-1}(f_0(\pi_X(W)))$$

which is open because both f_0 and π_X are open and π_Y is continuous.

A bijective, continuous, open map is a homeomorphism. ∎

Proposition 4.1.39
Let X be a topological space with open cover $X = \bigcup_{i \in I} U_i$. If every U_i is Kolmogorov with respect to the subspace topology then so is X.

Proof. Consider $x, y \in X$. We may assume wlog that $x, y \in U_i$ for some i (otherwise we would be done). Then by assumption there exists $V \subset X$ such that $x \in U_i \cap V$ and $y \notin U_i \cap V$, whence $x \in V$ and $y \notin V$ as required. ∎

4.1.6 Symmetric Spaces

Definition 4.1.40
*Let X be a topological space. We say that x is **quasi-closed** if $y \in \overline{\{x\}} \implies x \in \overline{\{y\}}$.*

Proposition 4.1.41 (Symmetric Topology)
Let X be a topological space. Then the following are equivalent

a) For all $x \in X$ we have

$$\overline{\{x\}} = \{y \in X \mid y \text{ topologically indistinguishable from } x\},$$

*b) For all $x, y \in X$ $\quad x \in \overline{\{y\}} \implies y \in \overline{\{x\}}$ (i.e. every point is **quasi-closed**),*
c) Every open neighbourhood U of x contains $\overline{\{x\}}$,
d) X is partitioned by sets of the form $\overline{\{x\}}$, indexed by X_0.

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*In this case we say that X is **symmetric** or R_0. Further $x \in \overline{\{y\}}$ if and only if x and y become equal in X_0.*

Proposition 4.1.42
Let X be a symmetric topological space and $x, y \in X$. Then the following are equivalent

a) x and y are topologically indistinguishable,
b) $x \in \overline{\{y\}}$,
c) $y \in \overline{\{x\}}$,
d) $\overline{\{x\}} = \overline{\{y\}}$,
e) x and y become equal in the Kolmogorov Quotient X_0

Proposition 4.1.43 (Open Subspace is Symmetric)
Let X be a symmetric space and $U \subset X$ an open subset. Then U is symmetric and for every $x \in U$ we have

$$\mathrm{cl}_X(\{x\}) = \mathrm{cl}_U(\{x\})$$

Proof. By (4.1.21) we have $\mathrm{cl}_U(\{x\}) = \mathrm{cl}_X(\{x\}) \cap U$, so these are equal by (4.1.41).c). We conclude that the sets $\overline{\{x\}}$ partition U so that U is also symmetric. ∎

Proposition 4.1.44 (Symmetric Open Cover $\implies$ Symmetric)
Let X be a topological space with open cover $\bigcup_{i=1}^n U_i$ such that U_i are symmetric in the subspace topology. Then X is symmetric.

Proof. Suppose $x \in \mathrm{cl}_X(\{y\})$ then by (4.1.43) we have $x \in \mathrm{cl}_{U_i}(\{y\})$, whence $y \in \mathrm{cl}_{U_i}(\{x\})$ by (4.1.41) and $y \in \mathrm{cl}_X(\{x\})$. Therefore by (4.1.41) we have X is symmetric. ∎

Proposition 4.1.45
Let X be a topological space. Then the following are equivalent

a) X is both Symmetric and Kolmogorov
b) Every point $x \in X$ is closed.

Corollary 4.1.46
Let X be a symmetric topological space. Then every point of X_0 is closed.

4.1.7 Hausdorff Spaces

Definition 4.1.47
*Let X be a topological space. We say that X is **Hausdorff** if for all distinct pairs $x, y \in X$ there exists open neighbourhoods U, V of x and y respectively such that $U \cap V = \varnothing$.*

*Observe that **Hausdorff** $\implies$ **Kolmogorov**, **Symmetric**, but not conversely (e.g. co-finite topology).*

Proposition 4.1.48
Let X be a Hausdorff topological space. Then X is both Kolmogorov and Symmetric. Further singletons are closed.

4.1.8 Convergent Sequences

Proposition 4.1.49 (Convergent Sequence)
Let X be a topological space, (x_n) a sequence in X and $x \in X$. Let $\mathcal{B}_x$ be a local base for x. Then the following are equivalent

a) For every $V \in \mathcal{U}_x$ there exists N such that

$$n \geq N \implies x_n \in V$$

b) For every $V \in \mathcal{B}_x$ there exists N such that

$$n \geq N \implies x_n \in V$$

c) The function

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\to X \\ n &\to x_n \\ \infty &\to x \end{aligned}$$

is continuous at ∞ with respect to the topology on $\mathbb{N} \cup \{\infty\}$ given by open sets of the form $\{N, N+1, ...,\} \cup \{\infty\}$ and arbitrary subsets of $\mathbb{N}$.

In this case we say that $x_n \to x$.

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Proposition 4.1.50 (Continuous $\implies$ Sequentially Continuous)
Let $f : X \to Y$ be a continuous map and $x_n \to x$ a sequence in X. Then $f(x_n) \to f(x)$.

Proof. If we regard the sequence as a continuous map $\mathbb{N} \cup \infty$ (4.1.49) then the result follows from (4.1.26). ∎

Proposition 4.1.51 (Limits are Unique in Kolmogorov Spaces)
Let X be a Kolmogorov topological space. Then limits of sequences are unique.

Proposition 4.1.52 (Sequentially Continuous $\implies$ Continuous)
Let X, Y be topological spaces, $f : X \to Y$ a map and $x \in X$ a point with a countable *local base $\mathcal{B}_x$. Suppose that for all sequences $x_n \to x$ we have $f(x_n) \to f(x)$. Then f is continuous at x.*

In particular if X is first countable then $f : X \to Y$ is continuous iff it is sequentially continuous.

Proof. Consider a point $x \in X$ and $y \coloneqq f(x)$. By hypothesis $\mathcal{B}_x = \{U_n\}$ is countable. By replacing with the base $U_n' \coloneqq \bigcap_{k=1}^{n} U_k$ we may suppose that

$$U_1 \supseteq U_2 ... \supseteq U_n \supseteq ...$$

is decreasing. Suppose f is not continuous at x then by (4.1.25).d) there exists $V \in \mathcal{U}_{f(x)}$ such that $f(U_n) \setminus V \neq \varnothing$ for all n. Therefore we may choose $x_n \in U_n$ such that $f(x_n) \notin V$. Consider $U \in \mathcal{U}_x$ then by definition there is some N such that $U_N \subseteq U$. Furthermore $n \geq N \implies U_n \subseteq U_N \subseteq U \implies x_n \in U$. Therefore $x_n \to x$. By construction $f(x) \in V$ but $f(x_n) \notin V$ for all n and in particular $f(x_n) \nrightarrow f(x)$. ∎

4.1.9 Connected Topological Spaces

Proposition 4.1.53 (Connected Topological Space)
Let X be a topological space. Then the following are equivalent

a) X cannot be written as the disjoint union $U \sqcup V$ of two non-empty open subsets $U, V \subset X$
b) There are no proper non-empty subsets which are both open and closed

*In this case we say that X is **connected**.*

Proposition 4.1.54 (Connected Subset)
Let X be a topological space and $Z \subset X$. Then the following are equivalent

a) Z is connected in the subspace topology
b) Z cannot be written as a disjoint $(Z \cap U) \sqcup (Z \cap V)$ with both $Z \cap U, Z \cap V \neq \varnothing$

Proposition 4.1.55
Let X be a topological space and $\{Z_i\}_{i \in I}$ a family of connected subsets such that $\bigcap_{i \in I} Z_i \neq \varnothing$. Then $\bigcup_{i \in I} Z_i$ is connected.

Definition 4.1.56 (Connected Component)
*Let X be a topological space. We say $Z \subset X$ is a **connected component** if it is both connected and not properly contained in a connected subset.*

Proposition 4.1.57
Let X be a topological space. Then every connected subset Z is contained in a unique connected component.

Proof. Define

$$\overline{Z} \coloneqq \bigcup \{Z' \supset Z \mid Z' \subset X \text{ connected}\}$$

then by (4.1.55) this is connected. If $Z'' \supset \overline{Z} \supset Z$ is connected then by construction $Z'' \subset \overline{Z}$, so by definition $\overline{Z}$ is a connected component. Let $\overline{Z}'$ be another connected component containing Z, then by the same argument $\overline{Z}' \subset \overline{Z}$ and by definition $\overline{Z}' = \overline{Z}$. ∎

Proposition 4.1.58
Let X be a topological space, then the connected components partition X.

We denote this collection by $\pi_0(X)$.

Proof. Evidently $\{x\}$ is connected so that the connected components cover X by (4.1.57).

Suppose we have two connected components Z_1, Z_2 are not disjoint. Then by (4.1.55) $Z_1 \cup Z_2$ is connected. By maximality we have $Z_1 \cup Z_2 = Z_1 = Z_2$ as required. ∎

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4.1.10 Irreducible Topological Spaces

References:

- [Bou98b] (Chapter II §4.2)
- [Stacks] (Tag 004U)

Proposition 4.1.59 (Irreducible space)
Let X be a topological space. Then the following are equivalent

a) $X = Z_1 \cup Z_2$ closed implies either $Z_1 = X$ or $Z_2 = X$
b) $U, V \neq \varnothing \implies U \cap V \neq \varnothing$ for open sets U, V
c) $U \neq \varnothing \implies \overline{U} = X$ i.e. every non-empty open set is dense

and we say X is irreducible.

Proof. *a)* $\implies$ *b)* Suppose U, V are open sets such that $U \cap V = \varnothing$. Then $X = (X \setminus U) \cup (X \setminus V)$. By hypothesis $X = (X \setminus U)$ or $X = (X \setminus V)$ whence either U or V is empty.

b) $\implies$ *a)* Similar.

c) $\iff$ *b)* Follows directly from (4.1.24) ■

Proposition 4.1.60 (Irreducible Subset)
Let $Y \subset X$ be a subset of a topological space. Then the following conditions on Y are equivalent

a) Y is irreducible in the subspace topology.
b) $Y \subseteq Z_1 \cup Z_2 \implies Y \subseteq Z_1$ or $Y \subseteq Z_2$ where Z_1, Z_2 are closed subsets of X
c) $U \cap Y \neq \varnothing, V \cap Y \neq \varnothing \implies (U \cap V) \cap Y \neq \varnothing$ for U, V open

*and we say Y is an **irreducible subset**.*

Proof. *a)* $\implies$ *b)*. Suppose that Y is irreducible in the subspace topology and $Y \subseteq Z_1 \cup Z_2$. This implies $Y = (Z_1 \cap Y) \cup (Z_2 \cap Y)$ is a decomposition of closed sets. So either $Z_1 \cap Y = Y$ or $Z_2 \cap Y = Y \implies Y \subseteq Z_1$ or $Y \subseteq Z_2$ as required.

b) $\implies$ *a)*. Suppose that $Y = (Z_1 \cap Y) \cup (Z_2 \cap Y)$. Then $Y \subseteq Z_1 \cup Z_2$, and for example $Y \subseteq Z_1$, which implies $Z_1 \cap Y = Y$. ■

Proposition 4.1.61
*Let $Y \subset X$ be a **closed** subset then the following are equivalent*

a) Y is an irreducible subset
b) $Y = Z_1 \cup Z_2 \implies Y = Z_1$ or $Y = Z_2$ where Z_1, Z_2 are closed subsets of X
c) $Y \subseteq Z_1 \cup Z_2 \implies Y \subseteq Z_1$ or $Y \subseteq Z_2$ where Z_1, Z_2 are closed subsets of X

In other words in the lattice of closed subsets, the irreducible subsets are precisely the join-prime subsets.

Proof. *a)* $\implies$ *b)*. Clearly Z_1, Z_2 are also closed subsets of Y, so the result follows by definition.

b) $\iff$ *c)*. This is (2.4.2).

c) $\implies$ *a)*. This was already proven in (4.1.60). ■

Remark 4.1.62
Singletons $\{x\}$ are always irreducible.

Definition 4.1.63 (Irreducible Component)
We say that Y is an irreducible component if it is a maximal irreducible subset.

Proposition 4.1.64
Let $f : X \to Y$ be a continuous map and $Z \subset X$ irreducible. Then $f(Z)$ is irreducible in Y.

Proof. Suppose $f(Z) \subset Y_1 \cup Y_2$ for Y_1, Y_2 closed. Then $Z \subset f^{-1}(f(Z)) \subset f^{-1}(Y_1) \cup f^{-1}(Y_2)$. Then by (4.1.60).b) wlog $Z \subset f^{-1}(Y_1)$ therefore $f(Z) \subset f(f^{-1}(Y_1)) \subset Y_1$. Therefore by the same criterion $f(Z)$ is irreducible. ■

Proposition 4.1.65
Let $Y \subset X$. Then Y is irreducible iff $\overline{Y}$ is.

Proof. This follows from (4.1.60).b) and (4.1.19) that $Y \subset Z \iff \overline{Y} \subset Z$. ■

Proposition 4.1.66 (Decomposition into Irreducible Components)
A topological space X may be decomposed into irreducible components. More precisely

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a) Every irreducible component is closed
b) Every irreducible closed subset is contained in an irreducible component
c) X is the union of irreducible components

Proof. We prove each in turn

a) By (4.1.65) $\overline{Y}$ is irreducible and closed, so by maximality $Y = \overline{Y}$ is closed.
b) We may show that the lattice of closed subsets is chain complete so we may use to show that the lattice of irreducible closed subsets is also chain complete (2.4.4). Therefore we may apply (2.1.56) to find an irreducible component.
c) As $\{x\}$ is irreducible every element is contained in an irreducible component by b).

■

Corollary 4.1.67
For $x \in X$ the closure $\overline{\{x\}}$ is an irreducible closed subset.

Corollary 4.1.68
X is irreducible if and only if it has a single irreducible component.

Definition 4.1.69 (Generic Point)
Let Z be an irreducible closed subset of X, then we say $\eta \in X$ is a generic point of Z if $Z = \overline{\{\eta\}}$.

Proposition 4.1.70
Let X be an irreducible space then every open subset U is irreducible and connected.

Proof. By (4.1.59) U is dense in X. Therefore by (4.1.65) U is irreducible.

Suppose that U is not connected then by (4.1.54) $U = (U \cap V) \sqcup (U \cap W)$ with V, W open subsets of X and $U \cap V, U \cap W$ non-empty. Then as U is irreducible we have $U \cap V \cap W \neq \varnothing$, a contradiction. ■

Proposition 4.1.71
Let $X = \bigcup_{i=1}^{n} X_i$ where X_i are irreducible closed subsets, and mutually incomparable. Then the X_i are the irreducible components of X.

Proof. For any irreducible closed subset E we have by (4.1.61) $E \subset X_i$ for some i. In particular every irreducible component is contained in some X_i. On the other hand every X_j is contained in an irreducible component by (4.1.66). By incomparability we deduce that each X_i is an irreducible component, and that the set is complete. ■

Proposition 4.1.72
Let $Y \subset X$ such that $Y = \bigcup_{i=1}^{n} Y_i$ where Y_i are the irreducible components of Y. Then the set of irreducible components of $\overline{Y}$ is $\left\{\overline{Y_i}\right\}_{i=1\ldots n}$, and these are distinct.

Proof. By (4.1.65) $\overline{Y_i}$ are irreducible subsets which, by (4.1.23) cover $\overline{Y}$. By (4.1.66) Y_i is closed in Y and so by (4.1.21) $\overline{Y_i} \cap Y = Y_i$. This shows that the $\overline{Y_i}$ are distinct and also incomparable. Then (4.1.71) shows that these must be the irreducible components. ■

Proposition 4.1.73
Let $U \subset X$ be an open subset. There is an order isomorphism

$$\begin{aligned} \{Y \subset U \mid Y \text{ irreducible, closed and non-empty}\} &\longleftrightarrow \{Z \subset X \mid Z \text{ irreducible and closed and } Z \cap U \neq \varnothing\} \\ Y &\to \operatorname{cl}_X(Y) \\ Z \cap U &\leftarrow Z \end{aligned}$$

which restricts to irreducible components.

Proof. By (4.1.65) the first map is well-defined. Conversely $Z \cap U$ is closed in U and open in Z, so irreducible by (4.1.70).

By (4.1.59) $Z \cap U$ is dense in Z, that is $Z = \operatorname{cl}_Z(Z \cap U) \overset{(4.1.21)}{=} \operatorname{cl}_X(Z \cap U) \cap Z \implies Z \subset \operatorname{cl}_X(Z \cap U)$, and so by minimality they are equal.

By (4.1.21) $\operatorname{cl}_X(Y) \cap U = \operatorname{cl}_U(Y) = Y$. Therefore the maps are mutually inverse and order preserving. ■

Proposition 4.1.74
Suppose X is a topological space with open cover $\bigcup_{i \in I} U_i$ such that

a) U_i is irreducible for all $i \in I$ and

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b) $U_i \cap U_j$ is non-empty for all $i, j \in I$.

Then X is irreducible.

Proof. Suppose that $X = Z_1 \cup Z_2$. Pick some $i \in I$ then by (4.1.60) wlog $U_i \cap Z_1 \neq \varnothing$. Similarly for all $j \in J$ by assumption $U_i \cap U_j$ is a non-empty open subset of U_j and therefore dense (4.1.59). Further $U_i \cap U_j$ is contained in the closed subset $Z_1 \cap U_j$. We conclude that $U_j \subset Z_1 \cap U_j$. Therefore $X \subset Z_1$. By (4.1.60) we further conclude that X is irreducible. ∎

4.1.11 Noetherian Topological Spaces

Definition 4.1.75 (Noetherian)
*A topological space X is **Noetherian** if the lattice of closed subsets is Artinian (i.e. satisfies the descending chain condition).*

Proposition 4.1.76 (Decomposition into Irreducibles)
Let X be a Noetherian topological space. Then every closed subset Y may be expressed uniquely as a finite, incomparable union of irreducible closed subsets. These are precisely the irreducible components of Y.

In particular X has only finitely many irreducible components.

Proof. The lattice of closed subsets is distributive and Artinian by definition. Therefore the result follows from (4.1.61) and (2.4.7). ∎

Proposition 4.1.77
Let X be a Noetherian topological space and $Y \subset X$. Then Y is Noetherian.

Proposition 4.1.78
Let $X = \bigcup_{i=1}^{N} X_i$ be a topological space such that X_i is Noetherian under the subspace topology. Then X is Noetherian.

Proof. Let $\{Z_n\}_{n\in\mathbb{N}}$ be a descending chain of closed subsets of X. By definition $\{Z_n \cap X_i\}_{n\in\mathbb{N}}$ terminates for $n \geq m_i$. Then for $n \geq \max(m_1, ..., m_N)$ the sequence $\{Z_n\}_{n\in\mathbb{N}}$ also terminates. ∎

Proposition 4.1.79
Let X be a Noetherian topological space and $U \subset X$ an open subset. Then there is a bijection

$$\begin{aligned} \{Y \subset U \mid Y \text{ closed and non-empty}\} &\longleftrightarrow \{Z \subset X \mid Z \text{ closed and every irreducible component meets } U\} \\ Y &\to \ \mathrm{cl}_X(Y) \\ Z \cap U &\leftarrow \ Z \end{aligned}$$

which is order preserving and restricts to closed irreducible subsets and irreducible components.

Proof. By (4.1.77) and (4.1.76) $Y = Y_1 \cup ... \cup Y_n$ where Y_i are the irreducible components of Y. Then by (4.1.72) the irreducible components of $\overline{Y}$ are $\overline{Y_i}$, which evidently intersect U. Furthermore by definition $Z \cap U$ is closed in U and the maps are well-defined.

By (4.1.73) the maps restrict to closed irreducible subsets (resp. components) and are bijections, in particular for Z irreducible closed we have $\overline{Z \cap U} = Z$.

By (4.1.21) $\overline{Y} \cap U = Y$. For the case Z is closed then as before $Z = Z_1 \cup ... \cup Z_n$ is a decomposition into irreducible components. By assumption $Z_i \cap U \neq \varnothing$ for all $i = 1...n$. Therefore by the irreducible case (4.1.73) and (4.1.23)

$$\overline{Z \cap U} = \overline{Z_1 \cap U} \cup ... \cup \overline{Z_n \cap U} = Z_1 \cup ... \cup Z_n = Z.$$

Therefore the maps are mutually inverse as required. ∎

4.1.12 Krull Dimension

References:

- Éléments de géométrie algébrique IV [Gro64] (Chap. 0 §**14.4.1**)
- Some remarks on biequidimensionality of topological spaces and Noetherian schemes [Hei17]

For a Noetherian topological space X the closed subsets form an Artinian, distributive lattice. Furthermore the irreducible subsets are precisely the join-prime elements by (4.1.61). Therefore we may use the notions of Krull Lattice developed in Foundations.tex's Krull Dimension section ((2.5.2)).

Definition 4.1.80 (Chain of irreducibles)
*Let X be a Noetherian topological space. A **chain** of irreducible closed subsets*

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$$Z_0 \subsetneq Z_1 \subsetneq ... \subsetneq Z_n$$

is said to have ***length*** n*. A chain is* ***saturated*** *if there is no proper refinement, that is if* Y *is irreducible then*

$$Z_i \subseteq Y \subseteq Z_{i+1} \implies Y = Z_i \text{ or } Y = Z_{i+1}.$$

If in addition Z_n *(resp.* Z_0*) is maximal (resp. minimal) then the chain is* ***maximal****.*

Definition 4.1.81 (Krull Lattice of Closed Subsets)
Let X *be a topological space.*

- *The* ***Krull dimension*** $\dim(X)$ *of* X *is the maximal length of all chains of irreducible closed subsets. Note this may be* ∞*.*
- *The* ***height*** *or* ***codimension*** *of an irreducible closed subset* $Y \subseteq X$*, denoted* $\operatorname{codim}(Y, X)$*, is the maximal length of chains of irreducible subsets containing* Y*.*
- *When* Y *is not irreducible we write*

$$\operatorname{codim}(Y, X) = \inf_\alpha \operatorname{codim}(Y_\alpha, X)$$

where Y_α *varies among the irreducible components of* Y*.*

If $\dim(X) < \infty$ *then we say* X *is* ***finite-dimensional****. In this case it's clear the closed subsets of a topological space form a Krull Lattice where the irreducible closed subsets are precisely the join-prime elements of the lattice.*

Note any saturated chain for $Y \subset X$ *must start at* Y *and terminate at an irreducible component of* X*. In particular if* X *is irreducible then a saturated chain must terminate at* X*.*

Proposition 4.1.82 (Extending Chains)
Let X *be a finite-dimensional topological space. Then*

a) Every chain is contained in a saturated chain with the same endpoints
b) Every chain is contained in a maximal chain

Proposition 4.1.83 (Simple properties of (co-)dimension)
Let X *be a topological space. Then the following properties hold*

a) $\dim(X) = \sup_\alpha \dim(X_\alpha)$ *where* X_α *are the irreducible components of* X
b) $\operatorname{codim}(Y, X) = \sup_\alpha \operatorname{codim}(Y, X_\alpha)$ *where* X_α *are the irreducible components of* X *containing* Y
c) $\dim(Y) + \operatorname{codim}(Y, X) \leq \dim(X)$ *for every closed subset* $Y \subset X$
d) $\operatorname{codim}(Y, Z) + \operatorname{codim}(Z, T) \leq \operatorname{codim}(Y, T)$ *for* $Y \subset Z \subset T$ *irreducible closed subsets*
e) $Y \subsetneq Z$ *is a saturated chain of irreducible closed subsets if and only if* $\operatorname{codim}(Y, Z) = 1$*.*
f) Let Y *be a closed subset of* X*. Then* $\operatorname{codim}(Y, X) = 0 \iff$ *every irreducible component of* Y *is an irreducible component of* X*. In particular if* X *is irreducible then* $\operatorname{codim}(Y, X) = 0 \iff Y = X \iff \dim(X) = \dim(Y)$*.*

In particular if X *is finite-dimensional then all codimensions are also finite.*

Proposition 4.1.84
Let X *be an irreducible topological space and* $Y \subset X$ *a closed subset. Then* $\dim(Y) \leq \dim(X)$ *with equality iff* $Y = X$*.*

Proof. The inequality follows from (4.1.83).b). If $\dim(Y) = \dim(X)$ then by c) $\operatorname{codim}(Y, X) = 0$ and so by f) we have $Y = X$. ∎

Definition 4.1.85 (Properties)
Let X *be a topological space of finite dimension. Then we say* X *is*

- ***Equidimensional*** *if all irreducible components of* X *have the same dimension*
- ***Equicodimensional*** *if* $\operatorname{codim}(Y, X)$ *is constant as* Y *varies over minimal irreducible subsets of* X
- ***Biequidimensional*** *if all maximal chains of irreducible subsets have the same length.*
- ***Quasi-Biequidimensional*** *if every irreducible component is biequidimensional*
- ***Catenary*** *if any two saturated chains with the same endpoints, say* Y *and* Z*, have the same length, namely* $\operatorname{codim}(Y, Z)$

Observe that ***quasi-biequidimensional*** *+* ***equidimensional*** $\iff$ ***biequidimensional*** *and* ***irreducible*** $\implies$ ***equidimensional***

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Proposition 4.1.86 (Equivalent characterisations of biequidimensional)
Suppose X is a topological space of finite dimension. Then the following are equivalent

a) X is quasi-biequidimensional
b) X is catenary and every irreducible component is equicodimensional
c) X satisfies the codimension formula for $Z \subset Y$ irreducible

$$\dim(Y) = \dim(Z) + \operatorname{codim}(Z, Y)$$

d) X satisfies c) in the case $\operatorname{codim}(Z, Y) = 1$

Furthermore for irreducible subsets $Z \subset Y$

$$\operatorname{codim}(Z) = \operatorname{codim}(Z, Y) + \operatorname{codim}(Y)$$

Proof. This is a translation of (2.5.11) to the topological case. ∎

Proposition 4.1.87 (Codimension Formula)
Suppose X is a quasi-biequidimensional topological space. Then for every subset $Z \subset X$ which is irreducible and closed, and every closed subset $Y \subset Z$ the following relations hold.

$$\dim(Z) = \dim(Y) + \operatorname{codim}(Y, Z)$$
$$\operatorname{codim}(Y) = \operatorname{codim}(Y, Z) + \operatorname{codim}(Z)$$

Further every closed subset is also quasi-biequidimensional.

Proof. This follows from (2.5.13) and (2.5.15). ∎

Proposition 4.1.88
Suppose $X = \bigcup_{i \in I} U_i$. Then $\dim(X) = \sup_{i \in I} \dim(U_i)$.

Proof. By (4.1.73) a chain of irreducible closed subsets of U_i lifts to a chain of the same length in X. Therefore $\dim(U_i) \leq \dim(X)$ for all $i \in I$ whence $\sup_{i \in I} \dim(U_i) \leq \dim(X)$. Similarly given a chain of irreducible closed subsets

$$X_0 \subsetneq ... \subsetneq X_n \subseteq X$$

choose i such that $X_0 \cap U_i \neq \varnothing$. Then (4.1.79) shows that this restricts to a chain of the same length in U_i. Therefore $\dim(X) \leq \dim(U_i) \leq \sup_{i \in I} \dim(U_i)$. ∎

Proposition 4.1.89
Let X be a catenary topological space and $U \subset X$ an open subset. Then U is catenary.

Proof. Consider $Z \subset Y \subset U$ irreducible closed subsets. Then for any saturated chain

$$Z = Z_0 \subsetneq ... \subsetneq Z_n = Y$$

we have by (4.1.73)

$$\overline{Z} = \overline{Z_0} \subsetneq ... \subsetneq \overline{Z_n} = \overline{Y}$$

a saturated chain in X. As X is catenary we find $n = \operatorname{codim}\left(\overline{Z}, \overline{Y}\right)$. As the chain was arbitrary we find U is catenary. ∎

Lemma 4.1.90
Suppose that X is a topological space and $x \in X$ is quasi-closed. Then $\dim\left(\overline{\{x\}}\right) = 0$.

Conversely suppose Z is an irreducible closed set of dimension 0. Then $Z = \overline{\{z\}}$ for all $z \in Z$ and Z consists of quasi-closed elements.

Proof. By (4.1.67) $\overline{\{x\}}$ is irreducible. Suppose $\varnothing \neq Z \subset \overline{\{x\}}$ is closed then $y \in Z \Longrightarrow x \in \overline{\{y\}} \subset Z$ whence $\overline{\{x\}} = Z$. This shows that $\dim\left(\overline{\{x\}}\right) = 0$.

Conversely if $\dim(Z) = 0$ then $\overline{\{z\}} \subset Z$ is an irreducible subset for all $z \in Z$, whence they are equal. This also shows that z is quasi-closed. ∎

Proposition 4.1.91
Let X be a Noetherian topological space and Z a closed subset. Then

$$\dim(Z) = 0 \iff Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$$

for some $z_1, ..., z_n \in Z$ quasi-closed.

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Proof. By (4.1.76) $Z = Z_1 \cup ... \cup Z_n$ for Z_i is a decomposition into irreducible components. By (4.1.83).a) $\dim(Z_i) = 0$, so by (4.1.90) $Z_i = \overline{\{z_i\}}$.

Conversely $Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$ is a decomposition into irreducible closed subsets which are incomparable (4.1.90). By (4.1.71) these are the irreducible components. Therefore

$$\dim(Z) \overset{(4.1.83).a)}{=} \sup_i \dim\left(\overline{\{z_i\}}\right) \overset{(4.1.90)}{=} 0$$

■

Proposition 4.1.92 (Curves have co-finite topology)
Let X be an irreducible, Noetherian and symmetric topological space such that $\dim(X) = 1$. Then X_0 has the cofinite topology.

Proof. We may reduce to the case $X = X_0$ is Kolmogorov, then by (4.1.45) every point $x \in X$ is closed.

Suppose $Z \subset X$ is finite then $Z = \bigcup_{z \in Z} \{z\}$ is a finite union of closed subsets and therefore closed. Conversely if $Z \subset X$ is closed and proper then by (4.1.83).f) $\dim(Z) = 0$ and therefore Z is finite by (4.1.91). ■

Lemma 4.1.93 (Cofinite Topology is Quasi-Compact)
Let X be a set equipped with the cofinite topology. Then X is quasi-compact.

Proof. Let $\{U_i\}_{i \in I}$ be an open cover of X and choose any nonempty U_{i_0} from it. Then $X \setminus U_{i_0}$ is finite by definition of the cofinite topology, say $\{x_1, ..., x_n\}$. For each k choose $U_{i_k} \ni x_k$ from the cover. Then $\left\{U_{i_0}, U_{i_1}, ..., U_{i_n}\right\}$ is a finite subcover. ■

Proposition 4.1.94
Let X be a symmetric topological space. Then there is a correspondence

$$\begin{aligned} X_0 &\longleftrightarrow \{W \subset X \mid \dim(W) = 0 \text{ closed and irreducible}\} \\ x &\longrightarrow \overline{\{x\}} \end{aligned}$$

4.1.13 Product Topology

Definition 4.1.95 (Product Topology)
*Let $\{X_i\}_{i \in I}$ be a family of topological spaces. The **product topology** is the topology generated by the base of open sets of the form*

$$\prod_{i \in I} U_i$$

where $U_i \subset X_i$ is open and only finitely many are proper subsets of X.

Proposition 4.1.96
Let $\{X_i\}_{i \in I}$ be family of topological spaces. The family of sets given in (4.1.95) is a base. The topology generated is the smallest family of subsets for which the projection maps

$$\pi_i : \prod_{i \in I} X_i \to X_i$$

are continuous. Furthermore if each X_i is Hausdorff (resp. Kolmogorov), then so is the product.

Proposition 4.1.97 (Product of Sequences)
Let $X_1, ..., X_m$ be a family of topological spaces and $a_n^i \to \alpha^i$ for $i = 1...m$. Then $(a_n^1, ..., a_n^m) \to (\alpha^1, ..., \alpha^m)$.

Proposition 4.1.98 (Diagonal is Closed)
Let X be a topological space. Then X is Hausdorff if and only if $\{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Proposition 4.1.99
Let $f, g : X \to Y$ be continuous maps and Y a Hausdorff space. Then

$$\{x \in X \mid f(x) = g(x)\}$$

is closed.

Proof. The level set is $(f \times g)^{-1}(\Delta_Y)$ which by (4.1.98) is closed. ■

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4.1.14 Quasi-Compactness

Definition 4.1.100 (Cluster Point of a Sequence)
Let (x_n) be a sequence in a topological space X. We say that $x \in X$ is

*a) a **cluster point** (or **accumulation point**) of x_n if for every $U \in \mathcal{O}_x$ the set $\{m \in \mathbb{N} \mid x_m \in U\}$ is infinite*
*b) a **subsequential limit** of x_n if there exists some subsequence $x_{n_k} \to x$*

In general the latter implies the former, and if X is first countable the concepts are equivalent as we will see.

Lemma 4.1.101 (Cluster Point is Subsequential Limit Point)
Let X be a first countable topological space and x a cluster point of a sequence (x_n). Then there exists some subsequence x_{n_k} such that $x_{n_k} \to x$.

Proof. Let (U_n) be a local base at x. By replacing it with $U_n' \coloneqq U_1 \cap ... \cap U_n$ we may assume it is without loss of generality decreasing. Define $n_0 = 0$ and n_k inductively by

$$n_k \coloneqq \min\{m > n_{k-1} \mid x_m \in U_k\}$$

which is well-defined precisely because x is a cluster point. Then $x_{n_k} \in U_k$ which shows that $x_{n_k} \to x$ as required. ■

Definition 4.1.102
Let X be a topological space. We say it is

- ***Quasi-Compact** if every open cover of X has a finite subcover*
- ***Countably Quasi-Compact** if every* countable *open cover of X has a finite subcover*
- ***Sequentially Quasi-Compact** if every sequence (x_n) in X has a convergent subsequence*

If in addition X is Hausdorff (e.g. a metric space) we say that X is compact (resp. countably compact, sequentially compact).

Proposition 4.1.103 (Criteria for Countable Compactness)
Let X be a topological space. Then the following are equivalent

a) X is countably quasi-compact
b) Let $\{Z_n\}_{n\in\mathbb{N}}$ be a countable collection of closed subsets such that every finite set has non-empty intersection. Then the intersection $\bigcap_{n\in\mathbb{N}} Z_n$ is non-empty
c) For every increasing sequence of proper open subsets

$$U_1 \subseteq U_2 \subseteq ... \subseteq U_n \subseteq ...$$

the union $\bigcup_{n=1}^{\infty} U_n$ is proper.
d) For every decreasing sequence of non-empty closed subsets

$$Z_1 \supseteq Z_2 \supseteq ... \supseteq Z_n \supseteq ...$$

the intersection $\bigcap_{n=1}^{\infty} Z_n$ is non-empty

Proof. $a) \iff b)$ and $c) \iff d)$ are simple consequences of De Morgan's laws. For $a) \implies c)$ suppose $\bigcup_{n=1}^{\infty} U_n = X$ then by assumption $U_{n_1} \cup ... \cup U_{n_k} = X$, and by the increasing assumption $U_{n_k} = X$ which is a contradiction.

For $c) \implies a)$, suppose we have an open cover $X = \bigcup_{n=1}^{\infty} U_n$ define the increasing open cover

$$U_n' \coloneqq \bigcup_{i=1}^{n} U_i$$

by the contrapositive U_n' is not proper for some n, which means precisely that there is a finite subcover. ■

Proposition 4.1.104
Let X be a topological space. Then sequentially quasi-compact $\implies$ countably quasi-compact. Under the assumption of first countability the converse holds.

Proof. $\implies$) Consider a countable open cover $X = \bigcup_{n=1}^{\infty} U_n$ with no finite subcover. Then there exists $x_n \in X \setminus \bigcup_{i=1}^{n} U_i$. By assumption there is a convergent subsequence $x_{n_k} \to x$. We have $x \in U_N$ for some N but $n_k \geq N \implies x_{n_k} \notin U_N$.

$\impliedby$) Consider a sequence (x_n) and define

$$Z_n \coloneqq \overline{\{x_k : k > n\}}$$

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which is a decreasing sequence of non-empty closed subsets. By the countable compactness criteria (4.1.103) $Z := \bigcap_{n=1}^{\infty} Z_n$ is non-empty. However we may show this is precisely the set of cluster points of (x_n), i.e. there exists a cluster point $x \in Z$. By (4.1.101) there exists a subsequence such that $x_{n_k} \to x$ as required. ∎

Lemma 4.1.105 (Lindelof Lemma)
Let X be a second countable topological space. Then every open cover has a countable subcover.

Proof. Let $\mathcal{B}$ be a countable base and $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ an open cover of X. Note for every U_α we have some subfamily $\mathcal{B}_\alpha \subset \mathcal{B}$ such that $U_\alpha = \bigcup \mathcal{B}_\alpha$. Consider $\mathcal{B}' := \bigcup\{B_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathcal{B}$. By definition for every $V \in \mathcal{B}'$ we have $V \subset U_\alpha$ for some $\alpha \in \mathcal{A}$. In other words there exists a map $f : \mathcal{B}' \to \mathcal{A}$ such that $V \subset U_{f(V)}$. We may verify that $\{U_\alpha\}_{\alpha\in \mathrm{Im}(f)}$ is a countable subcover. ∎

We summarise the relationships between the concepts

Proposition 4.1.106
Let X be a topological space. Then

a) Quasi-Compact, Sequentially Quasi-Compact $\implies$ Countably Quasi-compact
b) First Countable and (Countably) Quasi-Compact $\implies$ Sequentially Quasi-Compact
c) Second Countable and Countably or Sequentially Quasi-Compact $\implies$ Quasi-Compact

Proof.
a) The first implication is obvious and the second implication is (4.1.104)
b) This is (4.1.104)
c) This follows from (4.1.105) and *a*)

∎

Proposition 4.1.107 (Compact subsets of Hausdorff Spaces are Closed)
Let X be a Hausdorff topological space and $Y \subset X$ a subset compact with respect to the subspace topology. Then Y is closed.

Proof. Fix $x \in X \setminus Y$. Then for every $y \in Y$ by assumption there exists disjoint open subsets $x \in U_y$ and $y \in V_y$. Then $\{V_y \cap Y \mid y \in Y\}$ is an open cover for Y, so there exists a finite set of points $y_1, ..., y_n$ such that $Y \subseteq V_{y_1} \cup ... \cup V_{y_n}$. Furthermore

$$U := U_{y_1} \cap ... \cap U_{y_n}$$

is an open neighbourhood of x disjoint from each V_{y_i} and therefore disjoint from Y. By (4.1.19) we see that $x \notin \overline{Y}$, which shows $Y = \overline{Y}$, i.e. Y is closed. ∎

Proposition 4.1.108 (Closed subsets of Compact Spaces are Compact)
Let X be a quasi-compact topological space and $Y \subset X$ a closed subset. Then Y is quasi-compact under the subspace topology.

Proof. Let $\mathcal{U} := \{U_\alpha\}_{\alpha\in\mathcal{A}}$ be an open cover for Y. Then by definition $U_\alpha = U'_\alpha \cap Y$, where $U'_\alpha \subset X$ are open. Evidently

$$\mathcal{U}' : \{U'_\alpha\}_{\alpha\in\mathcal{A}} \cup \{X \setminus Y\}$$

is an open cover of X. Therefore by assumption it has a finite subcover, which immediately yields a finite subcover of Y as required. ∎

Proposition 4.1.109 (Image of a Compact Space is Compact)
Let $f : X \to Y$ be a continuous map of topological spaces where X is quasi-compact. Then the image $f(X)$ is quasi-compact.

Further if Y is Hausdorff then f is an open map (i.e. $f(U)$ is open for every open $U \subset X$). If in addition f is injective then it is a homeomorphism onto its image.

Proof. We may assume without loss of generality that $f(X) = Y$. Let $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ be an open cover of Y. In other words $f(X) \subseteq \bigcup_{\alpha\in\mathcal{A}} U_\alpha \implies X \subseteq \bigcup_{\alpha\in\mathcal{A}} f^{-1}(U_\alpha)$. By assumption we have a finite subcover

$$X \subseteq f^{-1}\left(U_{\alpha_1}\right) \cup ... \cup f^{-1}\left(U_{\alpha_n}\right)$$

which implies

$$f(X) \subseteq U_{\alpha_1} \cup ... \cup U_{\alpha_n}$$

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since f is assumed to be surjective, and we see that $f(X)$ is compact. ∎

Analogously we have similar results for sequentially compact spaces, which may be conceptually simpler.

Proposition 4.1.110
Let X be a first countable Kolmogorov space and $Y \subset X$ a subset of a sequentially compact with respect to the subspace topology. Then Y is closed.

Proof. Consider $x \in \overline{Y}$ then by (4.1.19) there is some sequence $x_n \to x$. Further by assumption there is some subsequence $x_{n_k} \to y \in Y$. By uniqueness of limits (4.1.51) we have $x = y$, and in particular $\overline{Y} = Y$ is closed. ∎

Proposition 4.1.111
Let X be a sequentially compact topological space, and $Y \subset X$ a closed subset. Then Y is sequentially compact.

Proof. Let $x_n \in Y$ be a sequence. By assumption it has a convergent subsequence $x_{n_k} \to x$ where $x \in X$. By (4.1.18) x is an adherent point of Y and therefore lies in Y by (4.1.19). ∎

Proposition 4.1.112
Let $f : X \to Y$ be a continuous map of topological spaces where X is sequentially compact. Then the image $f(X)$ is sequentially compact.

Proof. Let $y_n = f(x_n)$ be a sequence in $f(X)$. Then by assumption $x_{n_k} \to x$, and therefore by (4.1.50) $y_{n_k} \to f(x)$. ∎

Proposition 4.1.113
Let X be a topological space with an open cover $X = \bigcup_{i=1}^{n} U_i$ such that U_i is quasi-compact under the subspace topology. Then X is quasi-compact.

Proof. Let $X = \bigcup_{\alpha \in \mathcal{A}} V_\alpha$ then $U_i = \bigcup_{\alpha \in \mathcal{A}} (V_\alpha \cap U_i)$. By assumption there is some finite subset $\mathcal{A}_i \subset \mathcal{A}$ such that $U_i = \bigcup_{\alpha \in \mathcal{A}_i} (V_\alpha \cap U_i)$. Then

$$X = \bigcup_{\alpha \in \mathcal{A}_1 \cup \ldots \cup \mathcal{A}_n} V_\alpha$$

Therefore X is quasi-compact. ∎

Proposition 4.1.114 (Noetherian $\iff$ Hereditarily Quasi-Compact)
Let X be a topological space. Then X is Noetherian if and only if every open subset is quasi-compact.

Proof. $\implies$) By (4.1.77) every open subset is Noetherian, so it is enough to show that X is quasi-compact. By definition the lattice of open subsets is Noetherian and by (2.1.63) this holds iff every family of open subsets has a maximal element.

Suppose that we have an open cover $X = \bigcup_{\alpha \in \mathcal{A}} U_\alpha$. Let $\mathcal{F}$ be the family of open subsets expressible as a finite union of sets U_α. By assumption it has a maximal element

$$U := U_{\alpha_1} \cup \ldots \cup U_{\alpha_n}$$

Suppose that $U \subsetneq X$ is proper then choose $x \in X \setminus U$. By hypothesis there is some α_{n+1} such that $x \in U_{\alpha_{n+1}}$. Then $U \subsetneq U \cup U_{\alpha_{n+1}} \in \mathcal{F}$, contradicting maximality. Therefore $U = X$ and X is quasi-compact as required.

$\impliedby$) Omitted. ∎

Proposition 4.1.115
Let X, Y be quasi-compact topological spaces. Then $X \times Y$ is quasi-compact.

Proof. Let $\{U_\lambda\}_{\lambda \in \Lambda}$ be an open cover of $X \times Y$. Therefore for each (x, y) there is a $\lambda(x, y)$ such that $(x, y) \in U_{\lambda(x,y)}$. By definition of the product topology (4.1.95) we may find open sets $V_{(x,y)} \subset X$ and $W_{(x,y)} \subset Y$ such that $(x, y) \in V_{(x,y)} \times W_{(x,y)} \subset U_{\lambda(x,y)}$.

Fix x then the family $\left\{W_{(x,y)}\right\}_{y \in Y}$ is an open cover of Y. So we may find a finite subset $J_x \subset Y$ such that $\left\{W_{(x,y)}\right\}_{y \in J_x}$ covers Y.

Consider the open sets

$$V'_x := \bigcap_{y \in J_x} V_{(x,y)}$$

which evidently cover X. Then there is a finite subset $I \subset X$ such that $\{V'_x\}_{x \in I}$ covers X.

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Then we claim that the family $\left\{U_{\lambda(x,y)}\right\}_{x\in I, y\in J_x}$ is a finite subcover of $X\times Y$. For given $(x',y')\in X\times Y$ there is an $x\in I$ such that $x'\in V_x'$, and $y\in J_x$ such that $y'\in W_{(x,y)}$. Evidently $x'\in V_{(x,y)}$, whence $(x',y')\in U_{\lambda(x,y)}$ as required.■

4.1.15 Connectedness

Definition 4.1.116
*We say a topological space X is **disconnected** if $X=U\cup V$ for two non-empty, disjoint open subsets U,V. Otherwise we say X is connected.*

Proposition 4.1.117
Let X be a topological space. The following are equivalent

a) X is connected
b) The only subsets which are both open and closed are X and $\varnothing$

Proposition 4.1.118
Let $f:X\to Y$ be a continuous map and X connected. Then $f(X)$ is also connected.

4.1.16 Order Topology

Definition 4.1.119 (Order Topology)
*Let $(X,\leq)$ be a totally ordered set. Then the **order topology** is the topology generated by the subbase of half-open intervals*

$$\{y\in X\mid y<x\}\text{ and }\{y\in X\mid y>x\}$$

for $x\in X$.

Proposition 4.1.120 (Base for order topology)
Let $(X,\leq)$ be a totally ordered set. Then the sets of the form

$$\{y\in X\mid y<x\},\{y\in X\mid y>x\}\text{ and }\{y\in X\mid x<y<z\}$$

for $x,z\in X$ constitute a base for the order topology.

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